

Research Statement

Jian Kang (jian.kang@rochester.edu)
University of Rochester

Graphs are ubiquitous in science and industry to model the increasingly interconnected world we are living in. For example, a molecule can be represented as a graph, where nodes are atoms and edges are chemical bonds; a social network is also a representative graph-structured data, where nodes are social media users and edges are their friendship relations. Despite its ubiquity, deep learning models designed for image or text might not directly work on graphs due to many fundamental challenges brought by graph-structure data. First, graph is a typical non-i.i.d. data, since the semantic meaning of a graph is determined not only by the node itself, but also its connections to other nodes in the graph. Second, graph connectivity among nodes makes graph topologically complex, since a node might be arbitrarily connected to other nodes, making the space of possible graphs exponentially large even with a fixed number of nodes. Third, modeling graphs is even more challenging due to the potential volatility of graphs, as real-world graphs such as online social media is rapidly increasing in size with new contents and users everyday. In light of these challenges, to understand the ever interconnected world, it is crucial to learn a graph learning model from a graph-structured data that performs well in a graph learning task. Achieving this goal requires careful design choices over three components: (M1) model that approximates the ideal world dynamics in an expressive way, (M2) task that models the problem we want the model to solve automatically, and (M3) data that offers useful experience and feedback from which a model will learn.

My research vision is to build a computational foundation of *learning the world with graphs*. In pursuit of this vision, my research focuses on three dimensions that correspond to three aforementioned key components (Figure 1).

D1. Model Trustworthiness. Despite the development of many expressive graph¹ learning models, a fundamental question largely remains nascent: how can we make graph learning *trustworthy*? It is crucial to propose a paradigm shift for graph learning, from answering *who* and *what* to answering *how* and *why*. Trustworthy graph learning could benefit a variety of real-world scenarios, including financial fraud detection, personalized medicine and disaster risk management. To study the trustworthiness of graph learning models, several key pillars need to be explored, including (P1) discerning that enhances the transparency, (P2) debiasing that makes the graph learning models free from bias and discrimination, and (P3) defending graph learning that improves the resilience against real-world noise and adversaries.

D2. Task Hardness. The majority of existing graph learning model mostly learn on static graphs in a transductive setting, i.e., all nodes, edges, and labels are in closed sets during training and testing. However, real-world graphs are often of high volatility, with new nodes, edges, and labels emerge everyday. For example, in every minute in 2023, there are over 360,000 tweets being sent on Twitter, and over 3,700 users download Instagram Threads (a social media platform by Meta). It is critical to develop theories, models, and systems that can adapt to the data dynamics over time and real-world open-endedness, so as to enhance the generalization of graph learning model at test time.

D3. Data Complexity. Graph is a succinct form to model relationships (i.e., edges) among entities (i.e., nodes). The current research landscape in graph learning often considers each entity in a graph atomic with a feature representation, which might not be the ideal way to model our world, since the world is becoming increasingly interconnected from micro to macro. Real-world graphs might be collected from different sources or at different granularities. For example, cells are made up of the connections among multiple molecules, which are in turn made up of the connections among atoms, forming a network of networks. A complex problem is often solved by the collective intelligence of a group of decision makers, where each decision maker can be regarded as a decision-making model, forming a network of models. Thus, a comprehensive understanding of the world shall acknowledge the complexity of real-world interconnectedness and calls for a rethinking on the role of graphs in learning, from graph as data that connects atomic entities to graph as context that connects datasets, modalities, or models.

Research Highlights. My research so far spans all dimensions and develops expressive and trustworthy graph learning models that learn from various types of graphs, including plain graph, temporal graph, and multimodal graph.

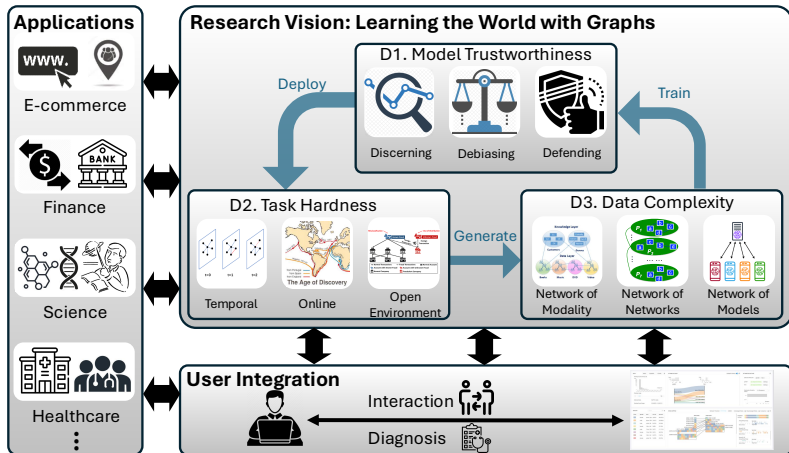


Figure 1: A research overview on learning the world with graphs.

¹We use the terms 'network' and 'graph' interchangeably.

For model trustworthiness, my research systematically studies principled solutions to discern, debias, and defend graph learning models. For task hardness, my research offers theoretical guarantees and empirically expressive models for temporal graph learning and online graph learning. For data complexity, my research introduces the first relational pretraining framework, which sheds lights on learning with multimodal graphs to further improve multimodal foundation models. In particular, my research highlights are summarized as follows.

- As a junior faculty, I am a Co-PI of a successful proposal to the Department of Homeland Security “Combating Dynamic and Networked Financial Crimes: Characterization, Monitoring, and Human-AI-Collaborative Visual Analytics”. I have also participated in several institution-level grant proposals, including a proposal to NSF AI Institute program to develop safe, transparent, responsible, open, neuro-inspired, and generative AI agent and a proposal to the NSF MRI programs for the acquisition of hardware for generative AI research in the Greater Rochester Area.
- I am recognized as one of the Rising Stars in Data Science by The University of Chicago in 2022 (Top 30 worldwide) and a Mavis Future Faculty Fellow by the University of Illinois Urbana-Champaign in 2022 (Top 4 in the Department of Computer Science at the University of Illinois Urbana-Champaign). I have also received the university nomination for 2022 Apple Scholars in AI/ML (Top 3 at the University of Illinois Urbana-Champaign).
- My research comprehensively studies learning with graphs in the wild from three dimensions: model trustworthiness, task hardness, and data complexity. My research has been key deliverables to multiple funded projects from NSF (FAI, MoDL SCALE) and DARPA (INCAS). Moreover, a variety of the developed models have been transferred to real-world products, such as core components in the recommender system of Reels (a short-form video platform by Meta) and in the ads ranking system at Meta.
- My research in the dimensions of model trustworthiness, task hardness, and data complexity results in over twenty publications in the prestigious conferences and journals (e.g., NeurIPS, ICLR, KDD, WWW, CIKM, TKDE, TVCG), with a spotlight in ICLR 2023 (Top 5% submissions).
- I am among the first to study fair graph learning and has been invited to deliver 14 invited talks, 6 guest lectures, and 2 keynote talks at the EAI-KDD workshop at KDD 2022 and at the GNN for the Sciences workshop hosted by The University of Chicago in 2024.
- My tutorials on fair graph learning and data quality aware graph learning as attracted over 300 audience in the top data mining conferences (CIKM 2024, SDM 2024, SDM 2023, KDD 2022, CIKM 2021).

1 Current Achievements

My current research contributions so far span all three dimensions of learning the world with graphs, boiling down to theoretical insights and principled solutions towards model trustworthiness, task hardness, and data complexity.

1.1 Dimension #1: Model Trustworthiness

Years of graph learning research have developed a plethora of theories and models that are successful in answering *what/who* types of questions. For example, *what* are the most relevant web pages with respect to a given query? *Who* can be grouped into the same community? Despite the prosperity of graph learning, a fundamental question largely remains nascent: how can we make graph learning model *trustworthy*? Along the dimension of model trustworthiness, it is crucial to propose a paradigm shift for graph learning, from answering *who* and *what* to answering *how* and *why* to understand the trustworthiness of graph learning models. Such a paradigm shift could benefit a variety of real-world scenarios, including financial fraud detection, personalized medicine, and disaster risk management.

Discerning Graph learning. Graph learning models are often opaque with respect to *how their predictions relate to the underlying graph structure*. In my research, we present a generic algorithmic framework named network derivative mining (N2N) to audit a graph learning model [7]. The key insights behind N2N is to construct a derivative network, each of whose edges quantitatively measure the influence as the rate of change (i.e., derivative) of a function over the learning results induced by the given graph learning algorithm. We further offer several instantiations of the proposed N2N framework, including PageRank, HITS ranking, spectral clustering, and matrix completion. For each instantiation, we develop efficient solution to construct the derivative network with linear complexity in both time and space. For PageRank, we further measure the node and subgraph influence as the aggregations of edge influence being incident to a node or induced in a subgraph. By our definition of influence, we instantiate N2N to a family of algorithms named AURORA that effectively identify influential graph elements with an $(1 - 1/e)$ approximation ratio due to the diminishing returns property of influence [8, 5]. Empirical results show that our proposed N2N framework can effectively identify influential graph elements.

Debiasing Graph learning. Though graph learning models achieve promising performance in many tasks, it is unclear *whether the graph learning results are fair, how to mitigate bias if not, and how would promoting fairness impact the model performance (e.g., classification accuracy)*. To answer these questions, we systematically study the individual fairness, degree fairness, and group fairness in graph learning.

Individual Fairness: We provide the first principled study (INFORM) in graph learning [6]. we first quantitatively measure the individual bias as the trace of a quadratic form of the learning results. Then we propose a family of three mutually complementary algorithms with respect to the input graph, the learning model, and the learning results, to mitigate our proposed individual bias measure. Furthermore, we prove that the cost of enforcing individual fairness relates to several factors of the graph structure: the number of nodes, the difference between the input graph and the similarity matrix, the rank and the largest singular value of the input graph. INFORM is further generalized to individually fair clustering on multi-view network (iFiG) by solving a multi-objective optimization problem [15]. Moreover, we define the ranking-based individual fairness named REDRESS to calibrate the distance comparison in INFORM with a plug-and-play debiasing module for any graph neural network [4].

Degree Fairness: We discover the theoretical and empirical insights into why graph neural networks exhibit performance disparity with respect to node degrees [13]. Theoretically, we prove that high-degree test nodes tend to have a lower misclassification probability regardless of how graph neural networks are trained. We further show that degree bias arises from various factors associated with the node degree (e.g., homophily of neighbors). Empirically, we show that degree bias are trivially limited by the expressive power of graph neural networks. To mitigate degree bias, we study why degree bias occurs after training by analyzing the gradients of model weights. Then we develop RAWLSGCN that normalizes the influence of node degrees on the gradients through doubly stochastic normalization on the input graph. Notably, RAWLSGCN is able to mitigate degree bias without modifying model architecture or introducing additional learnable parameters [11].

Group Fairness: We enforce the parity of a certain statistical measure among the pre-defined demographic groups. In my research, we empirically and theoretically prove that message passing amplifies the group bias when the numbers of neighbors from different demographic groups for each node are sufficiently large and unbalanced. Based on our analyses, we develop the model-agnostic BeMap that generates a fair neighborhood among different demographic groups using balance-aware sampling and performs message passing over the generated fair neighborhood [12]. Moreover, we study how to achieve group fairness among intersectional demographic groups defined by multiple sensitive attributes. Different from existing works that optimize a surrogate function of the statistical correlation between the sensitive attribute and the model prediction through adversarial learning, we develop INFOFAIR, which directly minimizes a variational representation of the mutual information between the model prediction and all sensitive attributes.

Interactive Visual Analytics: In my collaboration with researchers from Arizona State University, we develop an interactive visual analytics framework, named FAIR-RANKVIS, for end users to explore fairness in graph ranking. An overview of its interface is shown in Figure 2. The framework supports exploration of human-guided fairness definitions (i.e., group fairness, individual fairness) with respect to a human-defined combination of sensitive attributes and enables to diagnose the model trade-offs under different fairness definitions.

Defending Graph Learning. It aims to characterize the resilience of graph learning models against natural variations (e.g., noise, uncertainty) or adversarial attacks on graphs. Regarding robustness against natural variation on graphs, it remains nascent on the fundamental question like *how* uncertain is a graph learning model in its learning results? To answer this question, we study the post-hoc uncertainty quantification on graph neural networks. Regarding robustness against adversarial attacks, my research focuses on facilitating the deployment of fair graph learning in the real world via robustifying fair graph learning algorithms.

Uncertainty Quantification: The vast majority of existing works on Graph Convolutional Network (GCN) primarily focus on the accuracy while ignoring how confident or uncertain a GCN is on its predictions. The scarce existing efforts either fail to provide deterministic uncertainty estimate or have to change the training procedure of GCN. In my research, we propose JURYGCN, which is the first frequentist-based approach to quantify the uncertainty of GCN deterministically without modifying the GCN architecture or training procedure [10]. Its key idea is to quantify the uncertainty of a node as the width of confidence interval by jackknife. Efficient computation of the influence function is leveraged to estimate the change in GCN parameters without re-training in the jackknife estimator.

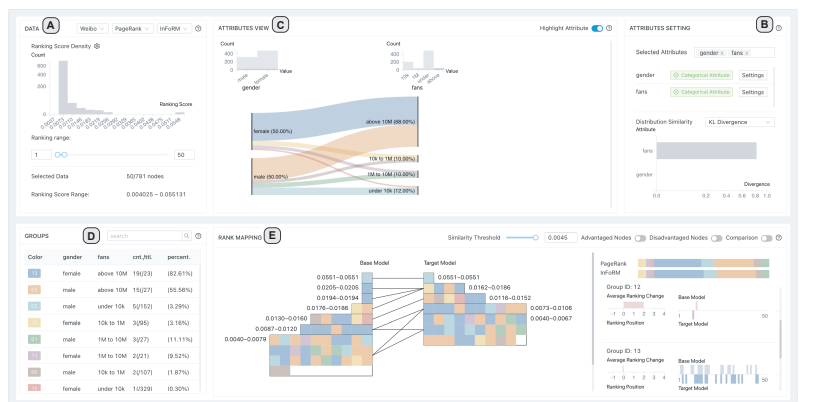


Figure 2: The user interface of FAIRRANKVIS.

Adversarial Fairness Attacks: Existing fair graph learning models assume the benevolence of input graphs and aim to mitigate the bias in the learning process. However, malicious activities in the real world are commonplace. It is critical to understand the resilience of a graph learning model with respect to adversarial attacks on fairness (i.e., fairness attacks). We formulate fairness attacks as a bi-level optimization problem, where the upper-level problem maximizes a bias function and the lower-level optimization problem minimizes a utility-centric loss function. We then develop a generic fairness attack framework named FATE, that poisons the input graph using the meta-gradient of the bias function. FATE is generalizable to attack any fairness definition on any graph learning model, which is evidenced by attacking group fairness and individual fairness of various graph neural networks.

Robust Fair Model Training: Enforcing individual fairness requires similarity measures in the input and output spaces, which are likely to be imperfect due to potential imprecise domain knowledge, noise, or adversaries. In my research, we aim to robustify individually fair learning model in the presence of imperfect similarity measure. We first quantitatively characterize the sensitivity of individual fairness with respect to the imperfect similarity measure. Based on that, we develop SOUL-A, which is a robust individually fair training framework by solving a minimax problem against L_∞ attack on the similarity measure. Our work could also pave the way for finding a more reliable similarity measure via the robustification of individually fair models.

1.2 Dimension #2: Task Hardness

Existing works in graph learning primarily focuses on transductive learning on static graphs. The dimension of task hardness aims to study graph learning in an open and dynamic environment, that is often harder due to the additional considerations of data dynamics over time and open-endedness in the real world. Along this dimension, we aim to develop graph learning models that (1) demonstrate strong generalization capability against distribution shifts and (2) accommodate the new data and labels that is missing historically.

Temporal Graph Learning. Temporal graph learning aims to capture node evolutionary patterns and accurately predict future links or node labels. The de facto standard for temporal graph learning is recurrent neural network (RNN) and self-attention mechanism (SAM). However, it is unclear whether RNN and SAM are indispensable to the success of temporal graph learning. In my research, we develop an empirically effective but conceptually and technically simple architecture named GraphMixer [3]. GraphMixer enjoys the simplicity of (1) neural architecture (multilayer perceptron and neighbor mean pooling only) and (2) input data (one-hop neighbors only and a fixed time-encoding function). Empirically, the simple input data format and neural architecture of GraphMixer lead to a smoother optimization landscape, a faster convergence speed, and a better generalization ability. Theoretically, we propose a unified theoretical framework toward the generalization error of RNN-based and SAM-based models. Guided by our analyses, we further simplify the temporal aggregation in GraphMixer and propose SToNe, that is purely based on stochastic gradient descent over a linear aggregation function [2]. SToNe not only enjoys a small theoretical generalization error but also bears a better overall empirical performance, a smaller model complexity, and an even simpler architecture compared to baseline methods including GraphMixer.

Online Graph Learning. Existing works in link prediction exclusively follow the conventional supervised learning setting, that (chronologically) creates the training and testing sets and trains a link predictor on the training set to predict links in the test set. However, this setting is often not realistic in real-world deployment, as the model fail to adapt over time to consistently meet the contexts and goals of the serving nodes. Thus, in my research, we re-formulate the link prediction task as a sequential decision-making problem. Each interaction of link prediction is considered as one round of decision-making in contextual bandits. Following this intuition, we propose PRB (PageRank Bandits) [1]. The key idea of PRB is in two-fold. First, we use two neural networks to estimate the exploitation score and exploration score of a node. Second, both scores are propagated via graph connectivity by PageRank. We theoretically show that the cumulative regret of PRB grows sublinearly. Moreover, PRB is extended to solve online node classification by introducing class-specific supernodes and predicting link between a node and a class-specific supernode.

1.3 Dimension #3: Data Complexity

The world is becoming increasingly interconnected at different granularities. Moreover, it is likely that real-world graphs might be collected from different sources. However, existing efforts in graph learning consider graph as data where nodes are atomic entities, without considering the potential internal structure of nodes. Along the dimension of data complexity, my research aims to fundamentally re-think the role of graph in learning. Specifically, I aim to achieve a paradigm shift from learning on data graphs to learning with contextual graphs, in which graph serves as a meta-structural context to connect entities, datasets, modalities, or even learning models.

Language-Audio Relational Pretraining. Playlist continuation, which extends a playlist by recommending songs in a personalized and musically cohesive manner, is a cornerstone in online music consumption. Existing playlist continuation models apply contrastive learning between text and audio only without considering the relationship between songs and the associated playlists. To learn better representations that encode playlist information, we propose LARP, a language-audio relational pretraining framework that injects graph connectivity information into the

training of audio encoder and text encoder. Central to the success of LARP is the introduction of playlist modality and the creation of multimodal graph that connects data in three modalities: text, audio, and playlist. A three-layered contrastive pretraining is developed to learn meaningful representation by contrasting (1) between text and audio, (2) between audio and audio, and (3) between audio and playlist. Our developed technique significantly outperforms existing approaches and can even recommend cold-start songs for playlist continuation.

2 Future Research Directions

In the future, I will further expand my research landscape in graph learning along the dimensions of model trustworthiness by understanding interplays among different aspects of trustworthiness, task hardness by studying open-environment graph learning, and data complexity by learning with contextual graphs.

Future Plan #1: (Model Trustworthiness) Interplays among Aspects of Trustworthy AI. Major efforts in trustworthy AI aim to guarantee one aspect of trustworthiness (e.g., transparency, fairness, robustness, privacy), while overlooking the inter-relationships among them. Without deep understanding on their implicit relationships, the trustworthy AI techniques would not be trustworthy and conscious enough to adapt the learned knowledge on unknown factors that do not exist previously. Thus, in the future, I am devoted to studying the interconnections among different aspects of trustworthy AI, which helps make an AI algorithm to be trustworthy in general rather than respecting to only one aspect. For example, robust model may help remove spurious correlation in model explanations and also ensure privacy; explainable model may help identify if a model has privacy leakage and if the algorithm is biased. I am also interested in exploring the potential applications of trustworthy AI, such as medical diagnosis and policy making.

Future Plan #2: (Task Hardness) Open-environment Graph Learning. An underlying assumption behind the majority of existing graph learning models is that all possible labels are known apriori in a closed label set during training, which we term as learning in a *closed-world environment*. Yet, graph learning models trained with this assumption might fail, especially in high-stake applications with emerging novel knowledge. For example, nearly 60% banks, fintechs, and credit unions faced over \$500K in direct fraud losses in 2023. Despite restrictive regulations and strong financial fraud detection models, novel financial crimes always emerge and remain undiscovered. Thus, it is critical to operationalize a graph learning model in an open environments by informing the model users when the model misclassifies data into one of the known classes or when it encounters data from novel classes. One possible solution might be managing the beliefs of model predictions using the subjective logic. Specifically, dissonance in subjective logic informs us the probability of a node being classified to more than one classes, which indicates potential misclassification; and vacuity informs us the disbelief of model predictions resulted from lack of knowledge.

Future Plan #3: (Task Hardness) Graph Foundation Model. Foundation model, such as GPT, has achieved remarkable success in natural language understanding and image generation. Despite the success in vision and language, state-of-the-art graph learning models are still shallow graph neural networks. In the future, I would like to explore (deep) generative foundation model for graphs. There are many fundamental research challenges in developing graph foundation model. For example, graphs from different domains (e.g., social network, molecular graphs) have different distributions. It is unclear whether there exists a foundation model that could learn the diverse data distributions imposed by graphs from different domains. Besides, the space of possible graph topologies grows quadratically in the number of nodes, let alone the generation of node/edge features. It essentially requires the graph foundation model to be scalable in understanding and generating large-scale realistic graphs. I envision that a graph foundation model might serve as a world model for our interconnected society to help understand infectious disease, sociodynamics, opinion dynamics, etc.

Future Plan #4: (Data Complexity) From Graph as Data to Graph as Context. My research on language-audio relational pretraining (LARP) demonstrate the importance of learning with the multimodal graph (i.e., network of modalities). In the future, I will continue investigating how to better model the world with graphs. I aim to transform the role of graph in graph learning, from data graph to contextual graph. There are several exciting directions. First, I plan to study learning with network of networks, which is essentially a graph of multiple graph datasets. A network of networks might be critical to model the biological system to better understand spatial biology and develop effective antimicrobial drugs to benefit human-being. Moreover, I will study learning with network of learning models, which has promising potentials in many real-world scenarios. First, it might help make an informed decision by the collective efforts of multiple machine learning models, e.g., a better rare disease diagnosis model that leverage diagnosis models for other diseases with similar symptoms. Additionally, it might also leverage knowledge from machine learning models for different tasks to develop a meta-cognitive model that adapts to new tasks.

My ultimate research goal is to discover theories and develop principled solutions for learning the world with graphs. In the meantime, I will also actively conduct high-impact interdisciplinary research.

Ultimate Research Goal: Learning the World with Graphs. For my ultimate research goal, I aim to develop a *computational foundation of learning the world with graphs*. Compared with current research landscape of graph learning, learning the world with graphs relies on the collective efforts in all three key dimensions in my research vision, which includes (1) a trustworthy graph foundation model that is transparent, fair, and robust, (2) a domain-invariant

lifelong graph learning model that adapts to new knowledge over time and across domains, and (3) a graph-centric modeling of the world that encodes data (e.g., network of networks), concepts (e.g., network of modalities), and context (e.g., network of learning models) to achieve human-level understanding and reasoning. Though being challenging, I believe such a computational foundation could help understand our world comprehensively, achieve human-level reasoning ability, and benefit human-being in the long run.

Interdisciplinary Research. Multi-disciplinary collaboration is one of the major driving forces behind many scientific studies. I am privileged to have enjoyable collaborations with researchers from different disciplines (e.g., finance, material science, biomedical engineering) [10, 3, 2, 1] and different areas in computer science (e.g., human-computer interaction) [8, 5, 14, 6, 16, 9]. These collaborations benefit me in not only broadening my vision and knowledge but also enhancing my research impact. For example, my research in uncertainty quantification has helped me obtaining research funding to study reliable financial crime identification to counter transnational criminal activities; my research in graph learning has benefited my collaborator in material science to design polymer membranes with efficient CO₂ separation ability; and my research in temporal link prediction has inspired my collaborator in biomedical engineering to understand and control genetic communication in bacterial communities. In the future, I look forward to working with researchers from different areas of computer science (e.g., human-computer interaction, theoretical computer science, etc.) and different disciplines (e.g., finance, healthcare, social science, etc.) for trustworthy graph learning.

References

- [1] BAN, Y., ZOU, J., LI, Z., QI, Y., FU, D., KANG, J., TONG, H., AND HE, J. Pagerank bandits for link prediction. In *NeurIPS* (2024).
- [2] CONG, W., KANG, J., TONG, H., AND MAHDAVI, M. On the generalization capability of temporal graph learning algorithms: Theoretical insights and a simpler method. *arXiv preprint arXiv:2402.16387* (2024).
- [3] CONG, W., ZHANG, S., KANG, J., YUAN, B., WU, H., ZHOU, X., TONG, H., AND MAHDAVI, M. Do we really need complicated model architectures for temporal networks? In *ICLR* (2023).
- [4] DONG, Y., KANG, J., TONG, H., AND LI, J. Individual fairness for graph neural networks: A ranking based approach. In *KDD* (2021), pp. 300–310.
- [5] KANG, J., FREITAS, S., YU, H., XIA, Y., CAO, N., AND TONG, H. X-rank: Explainable ranking in complex multi-layered networks. In *CIKM* (2018), pp. 1959–1962.
- [6] KANG, J., HE, J., MACIEJEWSKI, R., AND TONG, H. Inform: Individual fairness on graph mining. In *KDD* (2020), pp. 379–389.
- [7] KANG, J., AND TONG, H. N2n: Network derivative mining. In *CIKM* (2019), pp. 861–870.
- [8] KANG, J., WANG, M., CAO, N., XIA, Y., FAN, W., AND TONG, H. Aurora: Auditing pagerank on large graphs. In *IEEE Big Data* (2018), IEEE, pp. 713–722.
- [9] KANG, J., XIE, T., WU, X., MACIEJEWSKI, R., AND TONG, H. Infofair: Information-theoretic intersectional fairness. In *IEEE Big Data* (2022), IEEE.
- [10] KANG, J., ZHOU, Q., AND TONG, H. Jurygcn: Quantifying jackknife uncertainty on graph convolutional networks. In *KDD* (2022), pp. 742–752.
- [11] KANG, J., ZHU, Y., XIA, Y., LUO, J., AND TONG, H. Rawlsgcn: Towards rawlsian difference principle on graph convolutional network. In *WWW* (2022), pp. 1214–1225.
- [12] LIN, X., KANG, J., CONG, W., AND TONG, H. Bemap: Balanced message passing for fair graph neural network. In *LoG* (2023), PMLR, pp. 37–1.
- [13] SUBRAMONIAN, A., KANG, J., AND SUN, Y. Theoretical and empirical insights into the origins of degree bias in graph neural networks. In *NeurIPS* (2024).
- [14] WANG, M., KANG, J., NAN, C., XIA, Y., FAN, W., AND TONG, H. Graph ranking auditing: Problem definition and fast solutions. *IEEE TKDE* (2020).
- [15] WANG, Y., KANG, J., XIA, Y., LUO, J., AND TONG, H. ifig: Individually fair multi-view graph clustering. In *IEEE Big Data* (2022), IEEE.
- [16] XIE, T., MA, Y., KANG, J., TONG, H., AND MACIEJEWSKI, R. Fairrankvis: A visual analytics framework for exploring algorithmic fairness in graph mining models. *IEEE TVCG* (2021).